Chapter 5 Breakout (Pt. I)

STAT 4714

Questions

1. 50% of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is:

   (a) anywhere from 2 to 5;

   (b) fewer than 3.

$n = 6$ $\quad p = 0.50$ $\quad \Rightarrow$ Binomial distribution w/ binary outcomes (give 4 weeks v. don't)

(a) $P(2 \le X \le 5)$

$$= \sum_{x=2}^{5} b(x; n=6, p=0.5)$$

$$= \sum_{x=2}^{5} \binom{6}{x} (.5)^x (.5)^{6-x}$$

or $P(2 \le X \le 5) = P(X \le 5) - P(X \le 1)$

$= \text{pbinom}(5, 6, 0.5) - \text{pbinom}(1, 6, 0.5)$

$= 0.875$

(b) $P(X < 3) = P(X \le 2) = \text{pbinom}(2, 6, 0.5) = 0.3438$

2. One prominent physician claims that 70% of those with lung cancer are chain smokers. If this is true,

$p = 0.7$

(a) find the probability that of 10 such patients with lung cancer, fewer than half are chain smokers;

$n = 10$

(b) find the probability that of 20 such patients with lung cancer, fewer than half are chain smokers.

$n = 20$

binary outcomes = {chain smoker, not}

$\Downarrow$

Binomial

(a) $P(X < 5) = P(X \le 4) = \text{pbinom}(4, n=10, 0.7) = 0.0473$

Math sol:

$$P(X \le 4) = \sum_{x=0}^{4} b(x; n=10, p=0.7)$$

$$= \sum_{x=0}^{4} \binom{10}{x} (0.7)^x (0.3)^{10-x}$$

(b) $P(X < 10) = P(X \le 9)$

$$= \sum_{x=0}^{9} \binom{20}{x} (0.7)^x (0.3)^{20-x}$$

or

$$= \text{pbinom}(9, 20, 0.7) = 0.0171$$

$X \equiv$ # of rats w/ disease

$p = 1 - 0.6 = 0.4$

3. It is known that 60% of mice inoculated with a serum are protected from a certain disease. If 5 mice are inoculated, find the probability that

$n = 5$

   (a) none contracts the disease;
   (b) fewer than 2 contract the disease;
   (c) more than 3 contract the disease.

binary outcome = {protected, not protected}

$\Rightarrow$ Binomial distribution

$b(x; 5, 0.4)$

(a) $P(X=0) = \text{dbinom}(0, 5, 0.4) = 0.0778$

(b) $P(X<2) = P(X \le 1) = \text{pbinom}(1, 5, 0.4) = 0.3370$

(c) $P(X>3) = 1 - P(X \le 3) = 1 - \text{pbinom}(3, 5, 0.4) = 0.0870$

4. The probabilities are 0.4, 0.2, 0.3, and 0.1, respectively, that a delegate to a certain convention arrived by air, bus, car, or train. What is the probability that among 9 delegates randomly selected at this convention, 3 arrived by air, 3 arrived by bus, 1 arrived by car, and 2 arrived by train?

$$\left.\begin{array}{l}\text{multiple outcomes}\\ \text{observations are vectors}\end{array}\right\} \Rightarrow \text{multinomial}$$

$X_i \equiv$ # of delegates taking i^{th} mode of transportation $(i = 1, \dots, 4)$

$$m(x_1, \dots, x_4 ; n = 3+3+1+2, p_1 = 0.4, p_2 = 0.2, p_3 = 0.3, p_4 = 0.1)$$

$$= \binom{9}{x_1, x_2, x_3, x_4} (0.4)^{x_1} (0.2)^{x_2} (0.3)^{x_3} (0.1)^{x_4}$$

$$P(x_1 = 3, x_2 = 3, x_3 = 1, x_4 = 2)$$

$$= \text{dmultinom}(c(3,3,1,2), 9, c(0.4, 0.2, 0.3, 0.1))$$

$$= \underline{0.0077}$$